INCH-POUND
MIL-PRF-3098/131E
27 July 2005
SUPERSEDING
MIL-PRF-3098/131D
27 August 1997

PERFORMANCE SPECIFICATION SHEET

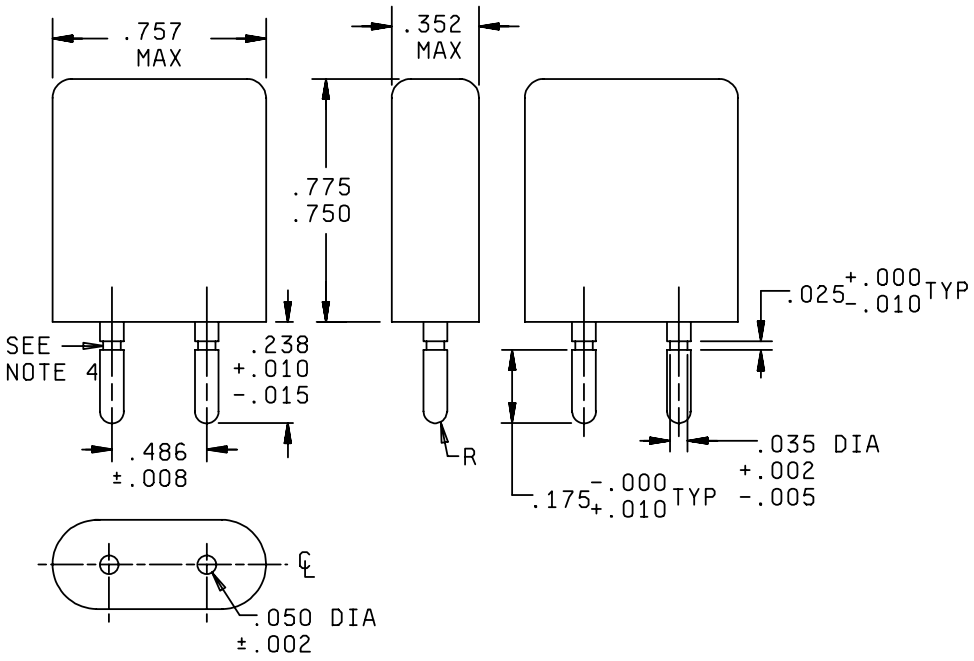
CRYSTAL UNIT, QUARTZ, CR150/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 45 MHz to 100 MHz; fifth mechanical overtone; controlled; series resonance.

Inches	mm
.002	.05
.005	.13
.008	.20
.010	.25
.025	.64
.035	.89
.050	1.27
.238	6.05
.352	8.94
.486	12.34
.750	19.05
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR150/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 45.0 MHz to 100 MHz, inclusive.

Operable temperature range: +55°C to +70°C, inclusive.

Operating temperature range (controlled): +70°C to +80°C, inclusive.

Frequency and equivalent resistance:

Equivalent resistance: 40 ohms, maximum.

Reference temperature: 75°C ±1°C.

Frequency tolerance: ±5 ppm.

Mode of oscillator: Fifth mechanical overtone.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 3.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Thermal shock:

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

NOTE: CF-150/U is similar to CR-143/U except for a wider frequency tolerance.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5955-0811-14)

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